

The empirical recurrence for OEIS A233641: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A233641 records “Empirical recurrence of order 66”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 729$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 66, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 66 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A233641 is “Number of $(n+1) \times (5+1)$ 0..2 arrays with every 2×2 subblock having the sum of the squares of the edge differences equal to 2.”. It has offset 1 and begins

4320, 99380, 2118568, 48384788, 1060656512, 24003655980, 531400908604, 11957878063196, ...

The entry states:

Empirical recurrence of order 66 (see link above).

As of the “Last modified” line on the live entry (Jan 01 2024 10:16:27, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 729$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 729$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1458$ exact terms of the model returns order 66 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	4	c_{18}	145312274921038095	c_{35}	149721748293374919114671	c_{52}	3082685
c_2	806	c_{19}	−136780023943868892	c_{36}	−854102389031837359350481	c_{53}	−718676
c_3	−1557	c_{20}	−2130041155501088316	c_{37}	−467225671514345981665792	c_{54}	−604501
c_4	−235326	c_{21}	1697524938961981282	c_{38}	1389133589927497264439247	c_{55}	18231
c_5	362466	c_{22}	24470476907499098448	c_{39}	1012679469461648351135425	c_{56}	614659
c_6	36013340	c_{23}	−15702096718836713042	c_{40}	−1669548510470307532980165	c_{57}	74572
c_7	−52203665	c_{24}	−221714941319301937627	c_{41}	−1573566930859440029210517	c_{58}	−27840
c_8	−3387763475	c_{25}	106033700159409569742	c_{42}	1413663700509295496163941	c_{59}	−5572
c_9	4762251826	c_{26}	1590055167749155461798	c_{43}	1759848078515478247566872	c_{60}	4555
c_{10}	213699329390	c_{27}	−491913309950016603514	c_{44}	−757946719243817838089409	c_{61}	1346
c_{11}	−289182199475	c_{28}	−9035929249612438530265	c_{45}	−1400807963932431870296982	c_{62}	−156
c_{12}	−9539262864742	c_{29}	1244081576924780283365	c_{46}	164328917228019590248204	c_{63}	−103
c_{13}	12212252262543	c_{30}	40630385556263498530251	c_{47}	774156125912300390802347	c_{64}	28
c_{14}	312384586267850	c_{31}	1400854920987453210384	c_{48}	84153071826021317569450	c_{65}	24
c_{15}	−370410405015124	c_{32}	−143949272540513232199849	c_{49}	−284376657622675397784234	c_{66}	−
c_{16}	−7696856553301130	c_{33}	−29584201619727395632866	c_{50}	−82915800309588933282780		
c_{17}	8249884297384563	c_{34}	398828868712782040142770	c_{51}	64010324617916721039170		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 66, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $66 + 4$ terms removed returns order 66 as well, so $\deg q_{\min} = 66 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $66 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 66 that this sequence satisfies — and that family turns out to have exactly one member.

References

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